Priya Patel

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Summary

Healthcare Data Scientist with 4+ years of experience building predictive models and statistical pipelines on clinical and Electronic Health Records(EHR) data. Experience applying machine learning (Python, scikit-learn, PyTorch) and statistical analysis to reproductive health, behavioral health, and omics domains. Proven ability to translate complex, high-dimensional data into evidence that drives clinical decisions. Skilled at handling sensitive patient data under strict HIPAA compliance and partnering with clinicians, policymakers, and operational leaders.

Experience

Data Scientist | BestSelf Behavioral Health — Oct 2024 - Present

- Built a Python-based predictive model to forecast client no-shows from EHR (Cerner) scheduling and demographic features, deployed to 69 programs, reducing missed appointments by 27% and directly improving patient access.
- Developed a real-time high-risk patient monitoring model integrating High-Risk flags from clinical notes and structured EHR data, enabling proactive care intervention for counselors and program directors across all clinics.
- Engineered an end-to-end claims denial prediction pipeline using SQL and Python, identifying payer-level denial patterns and contributing to a 12% improvement in revenue recovery.
- Build and maintain Power BI dashboards and SQL reports used by clinical and operational leaders to track patient access, service utilization, appointment attendance, and revenue cycle trends.
- Designed and validated statistical models for medication management compliance tracking, surfacing appointments outside two-week compliance windows for legal and compliance teams.
- Analyzed high-dimensional EHR datasets (patient access, demographics, service utilization) and prepared standardized reporting for state and county agencies on volume, performance, and outcomes.
- Collaborated with senior leadership, compliance, billing, and program directors to design KPIs and translate model outputs into actionable clinical and operational decisions

Data Scientist | University at Buffalo — Aug 2023 - Sep 2024

- Publication: SCREAM - Network Based Data Analysis in Omics Data
- Designed and implemented a deep learning integration framework (PyTorch) for clustering and analysis of 8,000 cell x 21,000-gene expression multiomics datasets
- Applied unsupervised Machine Learning(spectral clustering, latent representational learning) to reconstruct reduced feature spaces capturing underlying biological structure in combined scRNA-seq and scATAC-seq data using the SCREAM method.
- Built reproducible, modular pipelines using AnnData, Scanpy, NumPy, Pandas, and scikit-learn

Financial Data Analyst | Dark Horse Stocks — Jan 2022 - Aug 2023

- Built and maintained Power BI dashboards to track portfolio performance, market trends, and client insights.
- Automated data pipelines from AWS using Python and SQL, establishing reproducible data collection work flows for large-scale trend analysis.
- Maintained dynamic Power BI dashboard to produce weekly market insights and investment recommendations, contributing to an 8% increase in client engagement.

Data Analyst (Research Assistant Professor) | Saurashtra University — Jun 2019 - Jan 2022

- Utilized Alteryx's data-wrangling, data-cleaning, and modeling techniques to create a new database by merging data from various government databases using SQL for assessing female education policies impacting over 3,00,000 students.
- Applied hypothesis testing and spectral clustering to identify high-need subgroups in population data, delivering evidence-based recommendations to policymakers on resource allocation for 300,000+ individuals.

Skills

Programming: Python *(Scikit-learn, Pandas, NumPy, PyTorch, TensorFlow)*, MATLAB, R, PostgreSQL, SQL, Shell
Softwares and Tools: Power BI, Tableau, GCP BigQuery, Cloud SQL, GCP Looker, Git, Microsoft Excel, Microsoft PowerPoint
Data Science Skills: Predictive Modeling, Statistical Analysis, A/B Testing, High-Performance Computing, Database Management, Data Integration, CRM, Healthcare Data Analysis, Medicare Data, Survival Analysis, Data Visualization
Cloud Environments: Google Cloud, Microsoft Azure, AWS, Snowflake

Education

Master of Science in Data Science | SUNY, University at Buffalo	GPA: 4.00	Jun 2024
Master of Science in Applied Physics | The M. S. University of Baroda	GPA: 4.00	Jun 2019

Data Science & Machine Learning Projects

Network Based Data Analysis in Omics Data | University at Buffalo

- Applied advanced bioinformatics methods to address challenges in the analysis of high-dimensional omics data using AnnData and Scanpy libraries along with Numpy, Pandas and Scikit-learn.
- Employing latent representational learning to reconstruct reduced spaces capturing underlying data structures.
- Utilizing SCREAM method for clustering single-cell Multiomics data obtained from combined scRNA-seq and scATAC-seq.

Chronic Kidney Disease: A Data-Driven Diagnosis | University at Buffalo

- Preprocessed the raw data by handling missing values, feature reduction, and data normalization to ensure model performance.
- Implemented exploratory data analysis to gain insights into the dataset, identify trends & outliers, and select relevant features.
- Developed and implemented multiple statistical learning models, including logistic regression, support vector machines (SVM), decision trees, random forests, and neural networks, using Python and libraries such as scikit-learn and TensorFlow.
- Conducted a comparative analysis of different models' results, assessing performance and suitability for diagnosis.

Market Trend Analysis and Forecasting Using Statistical Models | University at Buffalo

- Conducted analysis of financial markets using SQL and Tableau to identify trends and economic indicators.
- Developed predictive models using techniques such as LSTM (Long Short-Term Memory) and ARIMA to forecast market trends and movements. And evaluated the models using metrics such as accuracy, precision, recall and F1-score.
- Implemented feature engineering strategies to enhance model performance by 23%.

Heart Bounding Box Prediction from X-Ray Images | University at Buffalo

- Developed a CNN model based on modified the ResNet-18 architecture that accommodates grayscale X-ray images as input and predicts four coordinates of bounding box corners, utilizing a dataset of 469 annotated images.
- Implemented various preprocessing and data augmentation techniques like random contrast changes, scaling, rotations, and translations to increase the size of the dataset, resulting in model robustness.
- Optimized computational efficiency by leveraging transfer learning with pre-trained weights for all layers except the first and last layers, thereby fine-tuning the model for heart detection.
- Attained an impressive prediction accuracy of approximately 95%, depicting the efficacy and reliability of the model.

Airport Database Management in SQL | University at Buffalo

- Handled data normalization utilizing BCNF analysis to enhance the structure and memory efficiency of the database.
- Executed multiple queries using PostgreSQL to extract valuable insights, contributing to informed decision-making processes.
- Adapted query optimization strategies to reduce computation time by 41%.
- Designed a interactive Power BI dashboard to visualize key metrics and trends for intuitive data exploration.

Certifications

Microsoft Certified: Azure Fundamentals
Deep Learning with PyTorch for Medical Image Analysis
Statistics with R Specialization
Data Visualization with Tableau
Database Management with PostgreSQL